

Commerce XI Maths App

Miscellaneous Exercise 1

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 1 - Sets and Relations | Total questions: 23

Question 1 - Q1(i)

Write the following set in set builder form.

$\{10, 20, 30, 40, 50\}$

Options

A. $\{x / x = 10n, n \in \mathbb{N}, n \leq 5\}$

B. $\{x / x = 10n, n \in \mathbb{W}, n < 5\}$

C. $\{x / x = 5n, n \in \mathbb{N}, n \leq 10\}$

D. $\{x / x = 10n, n \in \mathbb{N}, n < 5\}$

Correct answer: $\{x / x = 10n, n \in \mathbb{N}, n \leq 5\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{x / x = 10n, n \in \mathbb{N}, n \leq 5\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 2 - Q1(ii)

Write the following set in set builder form.

$\{a, e, i, o, u\}$

Options

A. $\{x / x \text{ is vowel of English alphabets}\}$

B. $\{x / x \text{ is consonant of English alphabets}\}$

C. $\{x / x \text{ is a letter of the word "vowel"}\}$

D. $\{x / x \text{ is an English alphabet}\}$

Correct answer: $\{x / x \text{ is vowel of English alphabets}\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{x / x \text{ is vowel of English alphabets}\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 3 - Q1(iii)

Write the following set in set builder form.

$\{\text{Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday}\}$

Options

A. $\{x / x \text{ represents day of a week}\}$

B. $\{x / x \text{ represents working day of a week}\}$

C. $\{x / x \text{ represents month of a year}\}$

D. $\{x / x \text{ represents weekend day}\}$

Correct answer: $\{x / x \text{ represents day of a week}\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{x / x \text{ represents day of a week}\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 4 - Q2(i)

If $U = \{x / x \in \mathbb{N}, 1 \leq x \leq 12\}$, $A = \{1, 4, 7, 10\}$, $B = \{2, 4, 6, 7, 11\}$, and $C = \{3, 5, 8, 9, 12\}$.

Write $A \cup B$.

Options

A. $\{1, 2, 4, 6, 7, 10, 11\}$

B. $\{4, 7\}$

C. $\{1, 10\}$

D. $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

Correct answer: $\{1, 2, 4, 6, 7, 10, 11\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{1, 2, 4, 6, 7, 10, 11\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 5 - Q2(ii)

If $U = \{x / x \in \mathbb{N}, 1 \leq x \leq 12\}$, $A = \{1, 4, 7, 10\}$, $B = \{2, 4, 6, 7, 11\}$, and $C = \{3, 5, 8, 9, 12\}$.

Write $B \cap C$.

Options

A. $\{\}$

B. $\{3, 5, 8, 9, 12\}$

C. $\{2, 4, 6, 7, 11\}$

D. $\{4, 7\}$

Correct answer: $\{\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 6 - Q2(iii)

If $U = \{x / x \in \mathbb{N}, 1 \leq x \leq 12\}$, $A = \{1, 4, 7, 10\}$, $B = \{2, 4, 6, 7, 11\}$, and $C = \{3, 5, 8, 9, 12\}$.

Write $A - B$.

Options

- A. $\{1, 10\}$
- B. $\{2, 6, 11\}$
- C. $\{4, 7\}$
- D. $\{1, 2, 6, 10, 11\}$

Correct answer: $\{1, 10\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{1, 10\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 7 - Q2(iv)

If $U = \{x / x \in \mathbb{N}, 1 \leq x \leq 12\}$, $A = \{1, 4, 7, 10\}$, $B = \{2, 4, 6, 7, 11\}$, and $C = \{3, 5, 8, 9, 12\}$.

Write $B - C$.

Options

- A. $\{2, 4, 6, 7, 11\}$
- B. $\{\}$
- C. $\{3, 5, 8, 9, 12\}$
- D. $\{2, 4, 6, 7, 11, 12\}$

Correct answer: $\{2, 4, 6, 7, 11\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{2, 4, 6, 7, 11\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 8 - Q2(v)

If $U = \{x / x \in \mathbb{N}, 1 \leq x \leq 12\}$, $A = \{1, 4, 7, 10\}$, $B = \{2, 4, 6, 7, 11\}$, and $C = \{3, 5, 8, 9, 12\}$.

Write $A \cup B \cup C$.

Options

- A. $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$
- B. $\{1, 2, 4, 6, 7, 10, 11\}$

C. {3, 4, 5, 7, 8, 9, 12}

D. {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11}

Correct answer: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}. The other choices omit a required element, include an invalid element, or use a different operation.

Question 9 - Q2(vi)

If $U = \{x / x \in \mathbb{N}, 1 \leq x \leq 12\}$, $A = \{1, 4, 7, 10\}$, $B = \{2, 4, 6, 7, 11\}$, and $C = \{3, 5, 8, 9, 12\}$.

Write $A \cap (B \cup C)$.

Options

A. {4, 7}

B. { }

C. {1, 4, 7, 10}

D. {1, 10}

Correct answer: {4, 7}

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives {4, 7}. The other choices omit a required element, include an invalid element, or use a different operation.

Question 10 - Q3

In a survey of 425 students in a school, it was found that 115 drink apple juice, 160 drink orange juice and 80 drink both apple as well as orange juice.

How many drink neither apple juice nor orange juice?

Correct answer: 230

Explanation and steps

Use $n(A \cup B) = n(A) + n(B) - n(A \cap B) = 115 + 160 - 80 = 195$. Therefore, neither = $425 - 195 = 230$.

Question 11 - Q4

In a school there are 20 teachers who teach Mathematics or Physics, of these, 12 teach Mathematics and 4 teach both Physics and Mathematics.

How many teachers teach Physics?

Correct answer: 12

Explanation and steps

Let M be Mathematics and P be Physics. From $n(M \cup P) = n(M) + n(P) - n(M \cap P)$: $20 = 12 + n(P) - 4$, hence $n(P) = 12$.

Question 12 - Q5(i) $A \times A$

If $A = \{1, 2, 3\}$ and $B = \{2, 4\}$.

State the elements of $A \times A$.

Options

A. $\{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$

B. $\{(1, 2), (1, 4), (2, 2), (2, 4), (3, 2), (3, 4)\}$

C. $\{(1, 1), (2, 2), (3, 3)\}$

D. $\{(1, 2), (2, 1), (2, 3), (3, 2)\}$

Correct answer: $\{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 13 - Q5(i) $A \times B$

If $A = \{1, 2, 3\}$ and $B = \{2, 4\}$.

State the elements of $A \times B$.

Options

A. $\{(1, 2), (1, 4), (2, 2), (2, 4), (3, 2), (3, 4)\}$

B. $\{(2, 1), (2, 2), (2, 3), (4, 1), (4, 2), (4, 3)\}$

C. $\{(1, 2), (2, 2), (3, 2)\}$

D. $\{(1, 4), (2, 4), (3, 4)\}$

Correct answer: $\{(1, 2), (1, 4), (2, 2), (2, 4), (3, 2), (3, 4)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(1, 2), (1, 4), (2, 2), (2, 4), (3, 2), (3, 4)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 14 - Q5(i) $B \times A$

If $A = \{1, 2, 3\}$ and $B = \{2, 4\}$.

State the elements of $B \times A$.

Options

- A. $\{(2, 1), (2, 2), (2, 3), (4, 1), (4, 2), (4, 3)\}$
B. $\{(1, 2), (1, 4), (2, 2), (2, 4), (3, 2), (3, 4)\}$
C. $\{(2, 1), (2, 2), (2, 3)\}$
D. $\{(4, 1), (4, 2), (4, 3)\}$

Correct answer: $\{(2, 1), (2, 2), (2, 3), (4, 1), (4, 2), (4, 3)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(2, 1), (2, 2), (2, 3), (4, 1), (4, 2), (4, 3)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 15 - Q5(i) $B \times B$

If $A = \{1, 2, 3\}$ and $B = \{2, 4\}$.

State the elements of $B \times B$.

Options

- A. $\{(2, 2), (2, 4), (4, 2), (4, 4)\}$
B. $\{(2, 4), (4, 2)\}$
C. $\{(2, 2), (4, 4)\}$
D. $\{(1, 2), (1, 4), (2, 2), (2, 4), (3, 2), (3, 4)\}$

Correct answer: $\{(2, 2), (2, 4), (4, 2), (4, 4)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(2, 2), (2, 4), (4, 2), (4, 4)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 16 - Q5(i) Intersection

If $A = \{1, 2, 3\}$ and $B = \{2, 4\}$.

Find $(A \times B) \cap (B \times A)$.

Options

- A. $\{(2, 2)\}$
B. $\{(1, 2), (2, 1)\}$
C. $\{(2, 4), (4, 2)\}$
D. $\{\}$

Correct answer: $\{(2, 2)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(2, 2)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 17 - Q5(ii)

If $A = \{-1, 1\}$, find $A \times A \times A$.

Options

- A. $\{(-1, -1, -1), (-1, -1, 1), (-1, 1, -1), (-1, 1, 1), (1, -1, -1), (1, -1, 1), (1, 1, -1), (1, 1, 1)\}$
- B. $\{(-1, -1), (-1, 1), (1, -1), (1, 1)\}$
- C. $\{(-1, -1, -1), (-1, 1, 1), (1, -1, 1), (1, 1, -1)\}$
- D. $\{(-1, -1, -1), (1, 1, 1)\}$

Correct answer: $\{(-1, -1, -1), (-1, -1, 1), (-1, 1, -1), (-1, 1, 1), (1, -1, -1), (1, -1, 1), (1, 1, -1), (1, 1, 1)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(-1, -1, -1), (-1, -1, 1), (-1, 1, -1), (-1, 1, 1), (1, -1, -1), (1, -1, 1), (1, 1, -1), (1, 1, 1)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 18 - Q6(i)

If $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$, consider $R_1 = \{(1, 4), (1, 5), (1, 6)\}$.

Which statement is correct?

Options

- A. R_1 is a relation from A to B
- B. R_1 is not a relation because 2 and 3 have no images
- C. R_1 is not a relation because 1 has three images
- D. R_1 is a relation from B to A

Correct answer: R_1 is a relation from A to B

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives R_1 is a relation from A to B. The other choices omit a required element, include an invalid element, or use a different operation.

Question 19 - Q6(ii)

If $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$, consider $R_2 = \{(1, 5), (2, 4), (3, 6)\}$.

Which statement is correct?

Options

- A. R_2 is a relation from A to B
- B. R_2 is not a relation because the pairs are not in $A \times B$
- C. R_2 is a relation from B to A
- D. R_2 is not a relation because each first component is different

Correct answer: R_2 is a relation from A to B

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives R_2 is a relation from A to B. The other choices omit a required element, include an invalid element, or use a different operation.

Question 20 - Q6(iii)

If $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$, consider $R_3 = \{(1, 4), (1, 5), (3, 6), (2, 6), (3, 4)\}$.

Which statement is correct?

Options

- A. R_3 is a relation from A to B
- B. R_3 is not a relation because some first components repeat
- C. R_3 is a relation from B to A
- D. R_3 is not a relation because 2 and 3 have the same image

Correct answer: R_3 is a relation from A to B

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives R_3 is a relation from A to B. The other choices omit a required element, include an invalid element, or use a different operation.

Question 21 - Q6(iv)

If $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$, consider $R_4 = \{(4, 2), (2, 6), (5, 1), (2, 4)\}$.

Which statement is correct?

Options

- A. R_4 is not a relation from A to B
- B. R_4 is a relation from A to B
- C. R_4 is empty relation from A to B
- D. R_4 is the universal relation from A to B

Correct answer: R_4 is not a relation from A to B

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives R_4 is not a relation from A to B. The other choices omit a required element, include an invalid element, or use a different operation.

Question 22 - Q7 Domain

Determine the domain of the following relation.

$$R = \{(a, b) / a \in \mathbb{N}, a < 5, b = 4\}$$

Options

- A. {1, 2, 3, 4}
- B. {0, 1, 2, 3, 4}
- C. {4}
- D. {1, 2, 3, 4, 5}

Correct answer: {1, 2, 3, 4}

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives {1, 2, 3, 4}. The other choices omit a required element, include an invalid element, or use a different operation.

Question 23 - Q7 Range

Determine the range of the following relation.

$$R = \{(a, b) / a \in \mathbb{N}, a < 5, b = 4\}$$

Options

- A. {4}
- B. {1, 2, 3, 4}
- C. {0, 1, 2, 3, 4}
- D. {1, 2, 3, 4, 5}

Correct answer: {4}

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives {4}. The other choices omit a required element, include an invalid element, or use a different operation.